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MATHEMATICS

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Students are encouraged to complement this study guide with lectures, laboratory practicals, recommended textbooks, and other approved academic resources for a more comprehensive understanding of the course.

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Study Guide

About This Study Guide

This Mathematics Study Guide is designed to assist students in understanding and applying fundamental mathematical concepts involving integers, number bases, modular arithmetic, standard form, approximation, indices, logarithms, equations, and variations.

The guide provides a structured approach to learning the major topics in the course. Each study session focuses on an important mathematical concept and provides an explanation of the skills students are expected to develop. Students are encouraged to study each section carefully, work through examples, practice mathematical operations, and apply the concepts to real-life and practical situations. The guide also provides opportunities for revision and examination preparation. Regular practice is essential because mathematical skills improve through understanding, calculation, and repeated application.

Course Overview / Course Aim / Course Objectives

This course introduces students to important mathematical concepts and develops their ability to perform calculations, solve problems, manipulate mathematical expressions, and apply mathematical principles to practical situations.

The course aims to help students:

- Understand and perform basic operations involving integers.
- Convert numbers from one base to another and understand number-base systems.
- Perform arithmetic operations in different number bases.
- Understand the concept and applications of modular arithmetic.
- Express numbers in standard form and approximate numbers appropriately.
- Apply the laws of indices to mathematical calculations.
- Understand logarithms and their relationship with indices.
- Use logarithms to solve problems involving multiplication, division, powers, and roots.
- Solve simple equations and problems involving variations.
- Change the subject of a formula correctly.
- Apply direct, inverse, joint, and partial variation to practical problems.
- Develop accuracy and confidence in mathematical calculations.
- Apply mathematical concepts to computer programming and everyday situations.
- Prepare effectively for tests and examinations.

Major Topics Covered

- Basic operations of integers.
- Number bases and conversion between bases.
- Decimal fractions in different bases.
- Arithmetic operations in number bases.
- Applications of number bases to computer programming.
- Modular arithmetic.

- Applications of modular arithmetic in daily life.
- Standard form.
- Approximation, decimal places, and significant figures.
- Laws of indices.
- Negative, zero, and fractional indices.
- Logarithms and logarithm tables.
- Calculations involving powers and roots.
- Relationship between indices and logarithms.
- Simple equations and change of subject of formulae.
- Direct, inverse, joint, and partial variations.
- Applications of variations.
- Revision and examination preparation.

Background to the Course

Mathematics is a fundamental subject concerned with numbers, quantities, structures, patterns, relationships, and logical reasoning. It provides the skills needed to solve problems in everyday life as well as in areas such as science, engineering, technology, economics, business, medicine, and computer science. The ability to work with integers and different number systems is important because numbers are used to represent quantities and perform calculations in many different contexts. Number bases, for example, are particularly important in computing because computers use binary and other number systems to represent and process information.

Mathematics also provides methods for dealing with very large and very small numbers. Standard form makes such numbers easier to write and calculate with, while approximation allows values to be represented to an appropriate degree of accuracy.

Indices and logarithms provide powerful methods for simplifying calculations involving powers, roots, growth, and other mathematical relationships. Similarly, equations and variations help students describe relationships between quantities and solve practical problems.

This study material introduces students to these concepts in a systematic manner. Students are expected to develop both conceptual understanding and computational skills through regular practice.

Course Study Sessions

Session 1: Basic Operations of Integers

Students will learn about integers and the basic operations of addition, subtraction, multiplication, and division. The session explains the rules for working with positive and negative integers and helps students understand how signs affect mathematical operations.

Students will practice performing calculations involving integers and applying the appropriate rules when combining positive and negative numbers. They will also solve simple problems involving more than one operation.

Session 2: Conversion Between Number Bases

Students will learn about number systems and the concept of a base. The session introduces the conversion of numbers from one base to base ten and the conversion of base-ten numbers into other bases.

Students will also learn how place values operate in different number systems. Emphasis will be placed on understanding the value of each digit according to its position and the base being used.

The session also introduces the conversion of decimal fractions, sometimes referred to as base fractions, from one base to base ten. Students will practice applying place-value principles to both whole numbers and fractional numbers.

Session 3: Conversion of Numbers from One Base to Another

Students will learn how to convert numbers directly from one base to another. The session explains methods that can be used to convert numbers through base ten and other appropriate procedures.

Students will practice converting whole numbers and fractional numbers between different bases. They will also learn how to check their answers to ensure that the converted values are correct.

Session 4: Arithmetic Operations in Number Bases

Students will learn how to perform addition, subtraction, multiplication, and division using numbers written in different bases.

The session explains how the rules of ordinary arithmetic can be applied while taking the value of the particular base into account. Students will practise carrying and borrowing in different bases and develop confidence in solving number-base calculations.

The session also introduces applications of number bases to computer programming. Students will learn why systems such as binary are important in computers and digital technology.

Session 5: Modular Arithmetic

Students will learn the concept of modular arithmetic and understand the meaning of a remainder when numbers are divided by a particular modulus.

The session introduces the notation used in modular arithmetic and explains how numbers can be grouped according to their remainders. Students will perform addition, subtraction, and multiplication using modular arithmetic. Applications to daily life will also be discussed. Students will examine situations involving cycles and repeating patterns, such as days of the week, clocks, calendars, and other systems that repeat after a fixed interval.

Session 6: Standard Form and Approximation

Students will learn how to express very large and very small numbers in standard form. The session explains the structure of standard form and how powers of ten are used to represent numbers efficiently.

Students will also study approximation and learn how to round numbers to a specified number of decimal places and significant figures.

The session emphasizes the importance of accuracy and helps students determine the appropriate level of approximation for a particular problem.

Session 7: Indices

Students will learn the laws of indices and how they are used to simplify mathematical expressions involving powers.

The session covers the multiplication and division of expressions involving indices, powers of powers, and other applications of the basic laws of indices.

Students will also study negative, zero, and fractional indices. They will learn how these forms of indices can be interpreted and used in mathematical calculations.

A review of the first half-term's work will be conducted at the end of the session, followed by a periodic test to assess students' understanding and progress.

Session 8: Introduction to Logarithms

Students will learn the meaning and basic principles of logarithms. The session introduces logarithms using base ten and explains their relationship with powers. Students will study simple examples such as the logarithms of 10, 100, 1,000, and other powers of ten. They will learn how to identify the logarithm of a number and understand the connection between logarithmic and exponential forms.

The session also introduces the use of logarithm tables for multiplication and division. Students will learn the basic procedures required to use logarithm tables accurately. Students will continue their study of logarithms by applying them to calculations involving powers and roots.

The session explains how logarithms can simplify calculations that would otherwise require difficult multiplication, division, or root extraction. Students will practice using logarithmic methods to solve mathematical problems. Students will also study the relationship between indices and logarithms and learn how expressions involving powers can be transformed into logarithmic form and vice versa.

Session 9: Simple Equations and Variations

Students will learn how to solve simple equations and manipulate mathematical formulae. The session focuses particularly on changing the subject of a formula.

Students will learn how to isolate a required variable while maintaining equality throughout the process. They will practice rearranging formulae involving addition, subtraction, multiplication, division, powers, and roots.

The session also introduces different types of variation, including:

- Direct variation.
- Inverse variation.
- Joint variation.
- Partial variation.

Students will learn how to identify the type of relationship between quantities and formulate equations that represent these relationships.

Session 10: Applications of Variations

Students will apply the principles of direct, inverse, joint, and partial variation to mathematical and real-life problems.

The session helps students identify variables that change in relation to one another and determine unknown quantities using appropriate variation equations.

Students will solve practical problems involving relationships between quantities and learn how to interpret the answers obtained.

The session also provides comprehensive revision of the work covered during the term. Students will review important concepts, practice examination-style questions, identify areas of difficulty, and prepare for the final examination.

Recommended Study Approach

Students should:

- Read each topic carefully before attempting exercises.
- Understand the rules and principles behind each mathematical operation.
- Memorize important laws and formulae while ensuring that they understand how they are applied.
- Practice calculations regularly.
- Show all necessary working when solving problems.
- Check signs carefully when working with positive and negative numbers.
- Practice converting numbers between different bases.
- Work through examples involving modular arithmetic.

- Practice rounding numbers to decimal places and significant figures.
- Apply the laws of indices to different types of expressions.
- Practice using logarithms and logarithm tables.
- Learn how to rearrange formulae systematically.
- Identify the correct type of variation before solving a variation problem.
- Relate mathematical concepts to everyday situations and computer applications.
- Review previous lessons regularly.
- Attempt past questions and examination-style exercises.
- Use revision periods to identify and correct areas of weakness.
- Prepare adequately before periodic tests and examinations.

Consistent practice, careful working, regular revision, and a clear understanding of mathematical principles will help students develop strong mathematical skills and prepare effectively for further studies and examinations.

Reference Book

New General Mathematics SSS 1

M. F. Macrae et al.

STUDY SESSION ONE: BASIC OPERATION OF INTEGERS

1.0 INTRODUCTION

This session provides readers with a detailed explanation of the fundamental concepts of integers and the basic operations performed on them. It introduces integers as a set of whole numbers that include positive numbers, negative numbers, and zero, and explains their importance in mathematics and everyday applications.

The session also explains the number line, positive and negative integers, absolute value, and the basic arithmetic operations of addition, subtraction, multiplication, and division. It further highlights the rules governing operations involving integers, including the use of signs and the order of operations. In addition, the session provides practical examples of how integers are applied in areas such as temperature, finance, measurement, profit and loss, and everyday calculations.

LEARNING OUTCOMES FOR THE STUDY SESSION 1

When you have studied this session, you should be able to:

- i. Define integers and identify the different types of integers.
- ii. Represent and compare integers using a number line.
- iii. Explain the concept of absolute value of an integer.
- iv. Perform addition and subtraction operations involving integers correctly.
- v. Apply the rules of signs when multiplying and dividing integers.
- vi. Solve simple mathematical problems involving the basic operations of integers.
- vii. Apply integer operations to practical situations and everyday problems.

1.1 ADDITION OF WHOLE NUMBERS

Examples:

1. Add the following numbers
4109, 39787, 1501 and 7865
2. A trader brought 13 dozens of oranges, 1 gross of apples and 9 scores of pineapple. How many fruits did she buy altogether?

Solution

$$\begin{array}{r} 1. \quad 4,109 \\ 39,787 \\ 1,501 \\ + 7,865 \\ \hline 53,262 \end{array}$$

2. 13 dozens of oranges = 13×12 oranges
= 156 oranges
1 gross of apples = 144 apples
9 scores of pineapples = 9×20
= 180 pineapples

Total number of fruit that she bought altogether = $156 + 144 + 180$

$$\begin{array}{r} 156 \\ 144 \\ +180 \\ \hline 480 \end{array}$$

EVALUATION

1. What is the sum of 6119, 19786 and 1999?
2. A school library has 3 gross of maths textbooks, 7 scores of english textbooks and 8 dozens of basic technology textbooks. How many books altogether are in the library?

SUBTRACTION OF WHOLE NUMBER

Examples:

1. Find the difference between 42006 and 7998.
2. A boy was sent on an errand to buy 3 dozens of milk at #680 per dozen and 2 packs of sugar which cost #150 per packet. How much will he collect if he was given #2400?

Solution

1. Required difference $42006 - 7998$

$$\begin{array}{r} 42006 \\ - 7998 \\ \hline 34008 \end{array}$$

2. Cost of 3 dozens of milk at #680 per dozens

$$= 3 \times \text{\#}680 = \text{\#}2,040$$

$$\text{Cost of 2 packs of sugar at \#150 per pack} = 2 \times \text{\#}150$$

$$= \text{\#}300$$

$$\text{Total cost of the item brought} = \text{\#}2,040 + \text{\#}300$$

$$= \text{\#}2,340$$

Hence,

$$\text{The change that the boy will collect} = \text{\#}2,400 - \text{\#}2,340$$

$$\text{\#}2,400$$

$$\text{\#}2,340$$

$$\text{\#} \underline{\hspace{1cm}} 60$$

EVALUATION

1. Subtract 449 from 1,001
2. The number of students in a school is 1,819. What is the number of boys in the school ,if the number of girls is 27?

MULTIPLICATION OF NUMBERS

Examples:

1. Find the product of 819 and 39
2. Evaluate 79×109

Solution

1. Required product 819×36

$$\begin{array}{r} 819 \\ \times 36 \\ \hline 4914 \\ 2457 \\ \hline 29,484 \end{array}$$

2. 79×109

$$\begin{array}{r} 109 \\ \times 79 \\ \hline 981 \\ 763 \\ \hline 8,611 \end{array}$$

EVALUATION

1. Evaluate 417×29
2. What is the product of 439 and 17?

DIVISION OF NUMBERS

Examples

1. Find the value of $6,513 \div 13$
2. Given that $19 \times y = 323$. Find the value of y
3. What is the quotient of $\frac{3,618}{9}$

Solution

1. $6,513 \div 13$

$$\begin{array}{r} 501 \\ 13 \overline{) 6,513} \\ \underline{65} \\ 1 \\ \underline{0} \\ 13 \\ \underline{13} \\ 0 \end{array}$$

2. $19 \times y = 323$

$$Y = \frac{323}{19}$$

$$Y = 19 \overline{) 323} \begin{array}{r} 17 \\ \underline{19} \\ 133 \\ \underline{133} \\ 0 \end{array}$$

$$Y = 17$$

EVALUATION

1. What is the value of P if $19 \times P = 3819$?
2. In an estate, 26 people are living in a house. If there are 26026 people living in the estate altogether. How many houses are in the estate?

GENERAL/REVISION EVALUATION

1. Find the sum of 62429, 325, 1426 and 98
2. Find the difference between 76211 and 8899
3. If $16 \times q = 40960$. find the value of q
4. In a village of 17598 people, 9998 are male. how many female are in the village?

READING ASSIGNMENT

NGM SSS1, review test 1 and 2 pages 3-4.

WEEKEND ASSIGNMENT

1. Add the following numbers : 719,35,608 (a) 5459 (b) 6469 (c) 7469 (d) 8489
2. Find the difference between 10001 and 799 (a) 8202 (b) 7202 (c) 9202 (d) 1002
3. What is the product of 56 and 415? (a) 23240 (b) 23250 (c) 33240 (d) 25340
4. Evaluate the quotient $\frac{414}{9}$ (a) 64 (b) 46 (c) 56 (d) 76

5. Given that $43 \times A = 43043$, find the value of A. (a) 11 (b) 101 (c) 1001 (d) 2463

THEORY

1. Mr. Ade's pay slip reads thus

Basic salary	#15,500
Transport	#9,900
House allowance	#4900
Medical allowance	#8750
Other allowance	#3,870
- a) What is the gross salary of Mr. Ade?
- b) If Mr. Ade pays #500 as tax and #650 for pension and also repay a loan of #3,250. how much is his net income?
2. (a) There are 37 students in each classroom in a certain school. If the school has 19 classrooms, how many students are there in the school?
- (b) Given that $17 \times z = 28985$ Find the value of Z

STUDY SESSION TWO: NUMBER BASE CONVERSIONS

2.0 INTRODUCTION

This session provides readers with a detailed explanation of **Number Base Conversions**, a fundamental concept in mathematics and computer science. It introduces the concept of number systems and explains how numbers can be represented in different bases, including decimal, binary, octal, and hexadecimal.

The session also explains the principles and procedures involved in converting numbers from one base to another. It covers common conversion methods such as decimal-to-binary, binary-to-decimal, binary-to-octal, binary-to-hexadecimal, and conversions involving octal and hexadecimal number systems.

In addition, the session highlights the importance of number base conversions in computing and digital technology, particularly in the representation and processing of data by computers. The systematic procedures for performing conversions are presented with suitable examples to improve understanding and accuracy.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- Define a number system and explain the concept of number bases.
- Identify and describe common number systems, including decimal, binary, octal, and hexadecimal.
- Explain the importance of number base systems in mathematics and computer science.
- Convert numbers from one base to another using appropriate conversion method
- Perform decimal-to-binary, binary-to-decimal, decimal-to-octal, and decimal-to-hexadecimal conversions.
- Convert between binary, octal, and hexadecimal number systems.
- Apply number base conversion techniques to solve basic mathematical and computing problems.

2.1 NUMBER BASE CONVERSIONS

People count in twos, fives, twenties etc. Also the days of the week can be counted in 24 hours. Generally people count in tens. The digits 0,1,2,3,4,5,6,7,8,9 are used to represent numbers. The place value of the digits is shown in the number. Example: 395:- 3 Hundreds, 9 Tens and 5 Units. i.e.

$$395_{10} = 3 \times 10^2 + 9 \times 10^1 + 5 \times 10^0.$$

Since the above number is based on the powers of ten, it is called the base ten number system i.e.
 $= 300 + 90 + 5.$

Also $4075 = 4$ Thousand 0 Hundred 7 Tens 5 Units i.e. $4 \times 10^3 + 0 \times 10^2 + 7 \times 10^1 + 5 \times 10^0$ Other Number systems are sometimes used.

Example: The base 8 system is based on the power of 8. For example: Expand 647_8 , 26523_8 , 101101_2 ,

(a) $645_7 = 6 \times 7^2 + 4 \times 7^1 + 5 \times 7^0 = 6 \times 49 + 4 \times 7 + 5 \times 1$

(b) $26523_7 = 2 \times 7^4 + 6 \times 7^3 + 5 \times 7^2 + 2 \times 7^1 + 3 \times 7^0$

(c) $101101_2 = 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

EVALUATION

Expand The Following

1. 735_8 2. 1010011_2

2.2 CONVERSION TO DENARY SCALE (BASE TEN)

When converting from other bases to base ten the number must be raised to the base and added.

Worked Examples:

Convert the following to base 10

(a) 27_8 (b) 11011_2

Solutions:

(a) $27_8 = 2 \times 8^1 + 7 \times 8^0 = 2 \times 8 + 7 \times 1 = 16 + 7 = 23$

(b) $11011_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$
 $= 16 + 8 + 0 + 2 + 1 = 27$

EVALUATION

Convert The Following To Base Ten:

(a) 101011_2 (b) 2120_3

2.3 CONVERSION FROM BASE TEN TO OTHER BASES

To change a number from base ten to another base

1. Divide the base ten numbers by the new base number;
2. Continue dividing until zero is reached;
3. Write down the remainder each time;
4. Start at the last remainder and read upwards to get the answer.

Worked Examples:

1. Convert 68_{10} to base 4.

2. Convert 129_{10} to base 2

Solutions:

1.

5	68	
5	13	R 3
5	2	R 3
	0	R 2

233_5

2	129	
2	64	R 1
2	32	R 0
2	16	R 0
2	8	R 0
2	4	R 0
2	2	R 0
2	1	R 0
	0	R 1

10000001_2

EVALUATION

1. Convert 568_{10} to base 8

2. Convert 100_{10} to base 2

Bicimals

Base ten fractions, or decimals, are based on negative powers of ten

$$\begin{array}{ccccc}
 6 \times 10^0 & 5 \times 10^{-1} & 8 \times 10^{-2} & 3 \times 10^{-3} & \\
 & \swarrow & \uparrow & \searrow & \\
 & \mathbf{6.583} & & &
 \end{array}$$

Similarly we can have base two fractions, **bicimals**, based on negative powers of two

$$\begin{array}{ccccccc}
 1 \times 2^0 & & 1 \times 2^{-1} & & 0 \times 2^{-2} & & 1 \times 2^{-3} \\
 & \swarrow & \uparrow & \searrow & & & \\
 & \mathbf{1.101} & & & & &
 \end{array}$$

To convert a bicimal to a decimal, first express each digit as a power of two, then change the powers to fractions. Study the example below

Example 1

Convert the following bicimals to decimals.

a. 1.101 b. 10.011 c. 110.11

a. $1.101 = 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$

$$= 1 + 1 \times \frac{1}{2} + 0 \times \frac{1}{2^2} + 1 \times \frac{1}{2^3}$$

$$= 1 + \frac{1}{2} + 0 + \frac{1}{8}$$

$$= 1 + 0.5 + 0 + 0.125$$

$$= 1.625$$

$$\text{b. } 10.011 = 1 + 2^1 + 0 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 2 + 0 + 0 \times \frac{1}{2} + 1 \times \frac{1}{2^2} + 1 \times \frac{1}{2^3}$$

$$= 2 + 0 + 0 + \frac{1}{4} + \frac{1}{8}$$

$$= 2 + 0.25 + 0.125$$

$$= 2.375$$

$$\text{c. } 110.11 = 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$$

$$= 4 + 2 + 0 + \frac{1}{2} + \frac{1}{4}$$

$$= 6 + 0.5 + 0.25$$

$$= 6.75$$

EVALUATION

Convert the following bicimals to base ten.

1. 10.0001

2. 10.01

3. 11.1

4. 0.001

2.4 Conversion of number from one base to another base

A number given in one base other than base ten can be converted to another base via base ten.

Example 1

Convert: (a) 1534_{six} to base eight

(b) $8A9F_{\text{sixteen}}$ to base eight.

Solution

(a) 1534_{six} to base eight

First convert 1534_{six} to base ten.

$$1534_{\text{six}} = 1 \times 6^3 + 5 \times 6^2 + 3 \times 6^1 + 4 \times 6^0$$

$$= 216 + 180 + 18 + 4$$

$$= 418_{\text{ten}}$$

Now convert 418_{ten} to base eight.

8	418	Remainders
8	52	2 ↑
8	6	4
	0	6

$$\text{i.e. } 418_{\text{ten}} = 642_{\text{eight}}$$

$$\text{Thus, } 1534_{\text{six}} = 642_{\text{eight}}$$

(b) $8A9F_{\text{sixteen}}$ to base eight

$$8A9F_{\text{sixteen}} = 8 \times 16^3 + 10 \times 16^2 + 9 \times 16^1 + 15 \times 16^0$$

$$= 32768 + 2560 + 144 + 15$$

$$= 35487_{\text{ten}}$$

Now convert 35487_{ten}

8	35487	Remainders
9	4435	7 ↑
8	554	3
8	69	2
8	8	5

$$\begin{array}{ccc} 8 & 1 & 0 \\ & 0 & 1 \end{array}$$

$$\begin{array}{lcl} \text{i.e. } 35487_{\text{ten}} & = & 105237_{\text{eight}} \\ \text{thus, } 8A9F_{\text{sixteen}} & = & 105237_{\text{eight}} \end{array}$$

Example 2

Determine the number bases x and y in the following simultaneous equations:

$$32_x - 12_y = 9_{\text{ten}} \text{ and } 23_x - 21_y = 4_{\text{ten}}$$

Solution

$$32_x - 12_y = 9_{\text{ten}} \quad (1)$$

$$23_x - 21_y = 4_{\text{ten}} \quad (2)$$

Change equation (1) to base ten as follows:

$$(3 \times x^1 + 2 \times x^0) - (1 \times y^1 + 2 \times y^0) = 9$$

$$3x + 2 - y - 2 = 9$$

$$3x - y = 9 \quad (1a)$$

Similarly, change equation (2) to base ten:

$$\text{i.e. } x - y = 1 \quad (2a)$$

subtracting equations (2a) from (1a):

$$2x = 8$$

$$x = 4$$

Substituting $x = 4$ in (2a)

$$4 - y = 1$$

$$4 - 1 = y$$

$$y = 3$$

Thus, $x = 4$ and $y = 3$.

EVALUATION

- If x represents a base number in the following equations, what is the value of x ?
 - $315_x - 223_x = 72_x$
 - $405_x + 43_{\text{eight}} = 184_{\text{ten}}$
- Convert each of the following to the base indicated:
 - 10401.11_{seven} to base eight
 - $4B3F_{\text{sixteen}}$ to base twelve

GENERAL EVALUATION

- Convert
 - 1785_{10} to base 7
 - Convert 2125_6 to base 10
- Determine the number bases x and y in the following simultaneous equations:
 - $31_x + 20_y = 23_{10}$
 $23_x - 11_y = 5_{10}$
 - $26_x - 34_y = 1000_2$
 $38_x - 21_y = 111_5$
- Find the value of Q if $(Q_4)^2 = 100100_2$

READING ASSIGNMENT

WEEKEND ASSIGNMENT

1. Express 342_6 as number in base 10 (a) 134 (b) 341 (c) 143
2. Change the number 10010_2 to base 10 (a) 1001 (b) 40 (c) 18
3. Express in base 2, 100_{10} (a) 100100 (b) 1100100 (c) 11001
4. Convert 120 base 10 to base 3 (a) 11110_3 (b) 1210_3 (c) 12110_3
5. Convert 25 base 10 to base 2 (a) 11001_2 (b) 1001_2 (c) 1100_2

THEORY

1. Convert 2364_7 to base 10
2. Convert 105_{10} to base 2

STUDY SESSION THREE: BINARY NUMBERS (BASE 2 NUMBERS)

INTRODUCTION

This session provides readers with a detailed explanation of the **Binary Number System**, a fundamental concept in mathematics and computer science. It introduces the concept of binary numbers and explains how numbers are represented using the base-2 number system.

The session explains the principles and characteristics of the binary number system, including its use of only two digits, **0 and 1**. It also discusses the significance of binary numbers in computing and digital technology, where data and instructions are represented and processed using binary values.

In addition, the session presents the procedures for performing basic binary operations and conversions. It covers important topics such as binary-to-decimal, decimal-to-binary, binary-to-octal, and binary-to-hexadecimal conversions. Suitable examples are provided to make the procedures easier to understand and apply.

The session also highlights the importance of binary numbers in computer systems, digital electronics, data representation, and information processing. The systematic methods for working with binary numbers are presented to improve understanding, accuracy, and problem-solving skills.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define the binary number system and explain the concept of a base-2 number system.
- ii. Identify and describe binary numbers and explain the significance of the digits 0 and 1.
- iii. Explain the importance of binary numbers in mathematics, computer science, and digital technology.
- iv. Explain the place-value system of binary numbers and determine the value represented by each binary digit.
- v. Convert binary numbers to decimal numbers using appropriate conversion methods.
- vi. Convert decimal numbers to binary numbers using repeated division and other suitable techniques.
- vii. Convert binary numbers to octal and hexadecimal numbers using appropriate grouping methods.
- viii. Perform basic binary arithmetic operations, including binary addition, subtraction, multiplication, and division.
- ix. Apply binary number techniques to solve basic mathematical and computing problems.

3.1 ADDITION IN BASE TWO

We can add binary numbers in the same way as we separate with ordinary base 10 numbers.

The identities to remember are:-

$0 + 0 = 0$, $0 + 1 = 1$, $1 + 0 = 1$, $1 + 1 = 10$, $1 + 1 + 1 = 11$, $1 + 1 + 1 + 1 = 100$

Worked Examples

Example 1

Simplify the following

1. $1110 + 1001$
2. $1111 + 1101 + 101$

Solutions:

$$\begin{array}{r} 1. \quad 1110 \\ + 1001 \\ \hline 10111 \end{array}$$

$$\begin{array}{r} 2. \quad 1111 \\ + 1101 \\ \hline 101 \\ \hline 100001 \end{array}$$

Note: 11 take 1 carry 1
10 take 0 carry 1
100 take 0 carry 10

Example 2

1. $11011_{\text{two}} + 1111_{\text{two}}$
2. $10011 + 1110$
3. $110111 + 11011 + 10111$

Solution

$$\begin{array}{r} 1. \quad 11011_{\text{two}} + 1111_{\text{two}} \\ \quad \quad \quad 11011 \\ \quad \quad \quad 1111 \\ \hline 101010 \end{array}$$

$$\begin{array}{r} 2. \quad 10011 + 1110 \\ \quad \quad \quad 10011 \\ \quad \quad \quad 1110 \\ \hline 100001 \end{array}$$

$$\begin{array}{r} 3. \quad 110111 + 11011 + 10111 \\ \quad \quad \quad 110111 \\ \quad \quad \quad 11011 \\ \quad \quad \quad 10111 \\ \hline 1101001 \end{array}$$

EVALUATION

1. Simplify the following; $1001 + 101 + 1111$
2. $10101 + 111$

3.2 SUBTRACTION IN BASE TWO

The identities to remember on subtraction are: $0 - 0 = 0$, $1 - 0 = 1$, $10 - 1 = 1$, $11 - 1 = 10$, $100 - 1 = 11$

Worked Examples

Simplify the following:-

- (a) $1110 - 1001$ (b) $101010 - 111$

Solutions:

$$\begin{array}{r} (a) \quad 1110 \\ - 1001 \\ \hline 101 \end{array}$$

$$\begin{array}{r} (b) \quad 101010 \\ - 111 \\ \hline 1110 \end{array}$$

Example 2

4. $1001_{\text{two}} - 111_{\text{two}}$
5. $10001 - 1111$
6. $11010_{\text{two}} - 1111_{\text{two}}$

Solution

1. $1001_{\text{two}} - 111_{\text{two}}$

$$\begin{array}{r} 1001 \\ - 111 \\ \hline 10 \end{array}$$

2. $10001 - 1111$

$$\begin{array}{r} 10001 \\ - 1111 \\ \hline 10_2 \end{array}$$

3. $11010_{\text{two}} - 1111_{\text{two}}$

$$\begin{array}{r} 11010 \\ - 1111 \\ \hline 1011 \end{array}$$

1st column: $1 - 1 = 0$

2nd column: $0 - 1$ is not possible,
so borrow 1 from the next
column to give 2 i.e. 10. Then
 $10 - 1 = 1$.

3rd column: $1 - 1 = 0$ **3.3 MULTIPLICATION AND DIVISION IN BASE TWO**

In multiplication, $0 \times 0 = 0$, $1 \times 0 = 0$, $1 \times 1 = 1$.

When there is long multiplication of binary numbers, the principle of addition can be used to derive the answer.
Under division, the principle of subtraction can be used.

Worked Examples:

1. 1110×111 2. $110 \div 10$

Solution:

$$\begin{array}{r} 1110 \\ \times 110 \\ \hline 0000 \\ 1110 \\ 1110 \\ \hline 1010100 \end{array}$$

$$\begin{array}{r} 11 \\ 10 \overline{) 110} \\ \underline{10} \\ 10 \\ \underline{00} \end{array}$$

Example 2

1. 101011×110
2. 11101×111

Solution

1. 101011×110

The working is shown below without explanation

$$\begin{array}{r} 101011 \\ \times 110 \\ \hline 000000 \\ 110101 \\ 110101 \\ \hline 100000010 \end{array}$$

2. 11101×111

$$\begin{array}{r} 11101 \\ \times 111 \\ \hline 11101 \\ 11101 \\ 11101 \\ \hline 11001011 \end{array}$$

Example 3

1. $101010 \div 111$ (base two)

$$\begin{array}{r} 110 \\ 111 \overline{) 101010} \\ \underline{111} \\ 111 \\ \underline{111} \\ 00 \\ \underline{00} \end{array}$$

2. Divide 1010.01_{two} by 11_{two} giving your answer to 3 places after the binary point.

$$\begin{array}{r} 11.011 \\ 11 \overline{) 1010.010} \\ \underline{11} \\ 100 \\ \underline{-11} \\ 101 \\ \underline{-11} \\ 100 \\ \underline{-100} \\ 11 \\ \underline{-11} \end{array}$$

EVALUATION

1. Evaluate $10111 \div 110$
2. Evaluate 10001×11
3. Evaluate $10001 - 1110$

GENERAL EVALUATION

1. Evaluate 111101×111
2. Evaluate $4020_5 \div 11_5$
3. $11001 + 1111$
4. $1101 - 111$
5.
$$\begin{array}{r} 1111 \\ \times 110 \\ \hline \end{array}$$

READING ASSIGNMENT

Essential Mathematics for SS 1 pages 54 – 55

WEEKEND ASSIGNMENT

1. Express 342_6 as a number in base 10. (a) 342 (b) 3420 (c) 134
2. Change the number 10010 to base 10 (a) 18 (b) 34 (c) 40
3. Express in base two the square of 11 (a) 1001 (b) 1010 (c) 1011
4. Find the value of $(101)^2$ in base two (a) 1010 (b) 1111 (c) 1001
5. Multiply 100001_2 by 11 (a) 1001 (b) 1100011 (c) 10111

THEORY

1. Convert the following to binary number
 - i. 10_{ten}
 - ii. $(10_{\text{ten}})^2$
3. Calculate $110_2 \times (1011_2 + 1001_2 - 101_2)$
4. Multiply 34_5 by 22_5 .

STUDY SESSION FOUR: MODULAR ARITHMETIC

INTRODUCTION

This session provides readers with a detailed explanation of **Modular Arithmetic**, a fundamental concept in mathematics and computer science. It introduces the concept of modular arithmetic and explains how numbers are handled using a fixed modulus.

The session explains the basic principles and properties of modular arithmetic, including **congruence, remainders, and modulo operations**. It also examines how modular arithmetic can be used to determine whether two numbers are congruent with respect to a given modulus.

In addition, the session covers important operations in modular arithmetic, including **modular addition, subtraction, multiplication, and exponentiation**. Suitable examples are provided to demonstrate the procedures involved and to improve understanding and accuracy.

The session also highlights the importance of modular arithmetic in various areas of mathematics and computer science, particularly in **cryptography, computer programming, coding systems, hashing, and digital security**. The systematic procedures for solving modular arithmetic problems are presented to develop students' mathematical reasoning and problem-solving skills.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define modular arithmetic and explain the concept of a modulus.
- ii. Explain the meaning of congruence and determine when two integers are congruent modulo a given number.
- iii. Identify and calculate remainders using the modulo operation.
- iv. Perform basic modular arithmetic operations, including addition, subtraction, multiplication, and division where applicable.
- v. Apply the properties of modular arithmetic to solve mathematical problems.
- vi. Solve modular equations using appropriate methods.
- vii. Explain the importance of modular arithmetic in mathematics, computer science, and digital technology.
- viii. Apply modular arithmetic techniques to solve basic problems involving patterns, cycles, and remainders.
- ix. Describe practical applications of modular arithmetic in areas such as cryptography, computer programming, and data security.

4.1 Modular Arithmetic

In the previous section, we discovered a new kind of arithmetic, where we add positive integers by roating in number cycle. This arithmetic is called modular arithmetic. In our example, we ignored multiples of 4 and concentrated on the remainders. In this case we say that the modulus is 4

For example,

$$5 = 1 \pmod{4}$$

Where mod 4 means with modulus 4 or modulo 4

Note that $9 \div 4 = 2$, remainder 1

And $45 \div 4 = 11$ remainder 1

We say that 9 and 45 are equal modulo 4,

i.e. $9 = 45 = 1 \pmod{4}$

Example 1

Reduce 55 to its simplest form:

d. Modulo 3

e. Modulo 4

f. Modulo 5

g. Modulo 6

a. $55 \div 3 = 18$, remainder 1

$$55 = 1 \pmod{3}$$

b. $55 \div 4 = 13$, remainder 3

$$55 = 3 \pmod{4}$$

c. $55 \div 5 = 11$, remainder 0

$$55 = 0 \pmod{5}$$

d. $55 \div 6 = 9$, remainder 1

$$55 = 1 \pmod{6}$$

EVALUATION

1. Write down the names of four markets in your locality which are held in rotation over 4* days.

4.2 Addition, Subtraction and Multiplication Operations in Module Arithmetic

Addition and Subtraction

The table below shows an addition table (mod 4) in which numbers 0, 1, 2 and 3 are added to themselves.

First number	Second number				
	$\oplus$	0	1	2	3
	1	0	1	2	3
	2	1	2	3	0
	3	2	3	0	1
	4	3	0	1	2

In the table, multiples of 4 are ignored and remainders are written down. For example $2 \oplus 3 = 5 = 1 \pmod{4}$ and $2 \oplus 2 = 4 = 0 \pmod{4}$. note that we often use the symbol $\oplus$ to show addition in modular arithmetic

Example 1

Find a. $0 \ominus 3 \pmod{4}$, b. $1 \ominus 2 \pmod{4}$

b. Start at 0 and move in an anticlockwise direction three places.

The result is 1.

$$\text{Therefore, } 0 \ominus 3 = 1 \pmod{4}$$

c. Start at 1 and move in an anticlockwise direction two places. The result is 3.

$$\text{Therefore, } 1 \ominus 2 = 3 \pmod{4}.$$

First Number	Second number				
	$\ominus$	0	1	2	3
	0	0			
	1	1	0	3	
	2	2	1		
	3	3			

Notice the importance here of stating which number comes first, e.g. $2 \ominus 1 \neq 1 \ominus 2$

Example 2

Add $39 \oplus 29 \pmod{6}$

Either

$$\begin{aligned} 39 \oplus 29 &= 68 \\ &= (6 \times 11 + 2) \\ &= 2 \pmod{6} \end{aligned}$$

Or, expressing both numbers in mod 6

$$\begin{aligned} 39 \oplus 29 &= (6 \times 6 + 3) + (6 \times 4 + 5) \\ &= (3 + 5) \pmod{6} \\ &= 8 \pmod{6} \\ &= 2 \pmod{6} \end{aligned}$$

4.3 Multiplication

Example 1

Evaluate the following, modulo 4,

b. $2 \otimes 2$ b. $3 \otimes 2$ c. $33 \otimes 9$

$$\begin{aligned} \text{a. } 2 \otimes 2 &= 4 \pmod{4} \\ \text{b. } 3 \otimes 2 &= 4 + 2 = 2 \pmod{4} \\ \text{c. } 33 \otimes 9 &= 297 = 4 \times 74 + 1 = 1 \pmod{4} \\ &\text{Or expressing both numbers in mod 4} \\ 33 \otimes 9 &= 1 \times 1 \pmod{4} \\ &= 1 \pmod{4} \end{aligned}$$

Example 2

Evaluate the following in the given modulo.

$$\begin{aligned} \text{a. } 16 \otimes 7 \pmod{5} &\qquad \text{b. } 18 \otimes 17 \pmod{3} \\ \text{a. } 16 \otimes 7 &= 112 \\ &= 22 \otimes 5 + 2 \\ &= 2 \pmod{5} \end{aligned}$$

or

$$\begin{aligned} 16 &= 15 + 1 = 1 \pmod{5} \\ 7 &= 5 + 2 = 2 \pmod{5} \\ 16 \otimes 7 &= 1 \otimes 2 \pmod{5} \\ &= 2 \pmod{5} \end{aligned}$$

- c. $18 \otimes 7 \pmod{3}$
 $18 = 0 \pmod{3}$
 $17 = 2 \pmod{3}$
 $18 \otimes 17 = 0 \otimes 2 \pmod{3}$
 $= 0 \pmod{3}$

In examples 1, 2, it can be seen that it is usually most convenient to convert the given numbers to their simplest form before calculation.

EVALUATION

- Find the following numbers in their simplest form, modulo 4.
 - 15
 - 102
 - 38
- Find the values in the moduli written beside them.
 - $16 \otimes 7 \pmod{5}$
 - $80 \otimes 29 \pmod{7}$
 - $21 \otimes 18 \pmod{10}$

GENERAL EVALUATION

- Complete the multiplication modulo 5

$\otimes$	0	1	2	3	4	5
0	0	0	0	0		
1	0					
2	0					
3	0		1			
4					1	
5	0				0	

- The shorter hand of a clock points a 10. What number did it point to 29 hours ago?
 - find the simplest positive form of $-29 \pmod{12}$
 - Calculate $10 - 29 \pmod{12}$

READING ASSIGNMENT

New General Mathematics for SS 1 Page 239 ex. 20c 1 – 10

WEEKEND ASSIGNMENT

Find the simplest form of the following in the given moduli.

- $-75 \pmod{7}$ A. 4 B. 2 C. 5 D. 7
- $-56 \pmod{13}$ A. 10 B. 5 C. 9 D. 12

Find the values in the moduli written beside them.

- $8 \otimes 25 \pmod{3}$ A. 2 B. 5 C. 9 D. 4
- $27 \otimes 4 \pmod{7}$ A. 7 B. 5 C. 1 D. 3
- $21 \otimes 65 \pmod{4}$ A. 1 B. 9 C. 4 D. 8

THEORY

1. Calculate the following

a. $42 \oplus 28 \pmod{8}$

b. $12 \oplus 9 \pmod{4}$

2. Complete the multiplication modulo 6

$\otimes$	2	3	4	5
2				
3		3	0	
4				
5		3	2	1

STUDY SESSION 5: STANDARD FORM AND APPROXIMATION

5.0 INTRODUCTION

This session provides readers with a detailed explanation of **Standard Form and Approximation**, important concepts in mathematics used to represent, simplify, and estimate numerical values. It introduces the concept of standard form and explains how very large and very small numbers can be expressed in a convenient and systematic way.

The session explains the principles and procedures involved in writing numbers in **standard form**, also known as scientific notation. It covers the use of powers of ten and explains how positive and negative indices are applied when representing large and small numbers.

In addition, the session introduces different methods of **approximation**, including rounding numbers to a specified number of decimal places and significant figures. Suitable examples are provided to demonstrate how numbers can be approximated while maintaining an appropriate level of accuracy.

The session also highlights the importance of standard form and approximation in mathematics, science, engineering, and everyday calculations. The systematic procedures for converting, rounding, and estimating numbers are presented to improve accuracy, efficiency, and problem-solving skills.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define standard form and explain its purpose in representing numbers.
- ii. Express large and small numbers in standard form using powers of ten.
- iii. Convert numbers from standard form to ordinary form and vice versa.
- iv. Explain the meaning of positive and negative indices when used in standard form.
- v. Define approximation and explain its importance in mathematical calculations.
- vi. Round numbers to a specified number of decimal places accurately.
- vii. Round numbers to a specified number of significant figures using appropriate rules.
- viii. Estimate and approximate numerical calculations to obtain reasonable answers.
- ix. Apply standard form and approximation techniques to solve basic mathematical and real-life problems.
- x. Recognize the importance of accuracy and appropriate approximation in mathematical and scientific calculations.

5.1 STANDARD FORM

A number written in the form of $A \times 10^n$, such that A is a number between 1 and 10 ($1 \leq A \leq 10$) and n is a whole number (integer) is said to be in standard form.

Examples: 2×10^6 , 7×10^{-3} , 2.5×10^4 , 8.6×10^{-9} etc.

Work Example:

1. Express the following numbers in standard form:

(a) 300000 (b) 55 (c) 2,300,000 (d) 720,000,000 (e) 9,400,000,000

Solution

a) $300,000 = 3.0 \times 100,000$

$$= 3.0 \times 10^5$$

b) $55 = 5.5 \times 10$

$$= 5.5 \times 10^1$$

c) $2,300,000 = 2.3 \times 1,000,000$

$$= 2.3 \times 10^6$$

d) $720,000,000 = 7.2 \times 100,000,000$

$$= 7.2 \times 10^8$$

e) $9,400,000,000 = 9.4 \times 1,000,000,000$

$$= 9.4 \times 10^9$$

2. Change the following from standard form to ordinary form:

a) $5.1 \times 10^7 = 5.1 \times 10,000,000$

$$= 51,000,000$$

b) $2.5 \times 10^6 = 2.5 \times 1,000,000$

$$= 2,500,000$$

c) $3.4 \times 10^1 = 3.4 \times 10 = 34$

d) $9.8 \times 10^5 = 9.8 \times 100,000 = 980,000$

e) $6 \times 10^8 = 6 \times 100,000,000 = 600,000,000$

Since decimal fraction can be expressed as power of 10, they can also be expressed in standard form as shown in the example below:

3. Express the following fractions in standard form

a) 0.0015 (b) 0.000026 (c) 0.000000067 (d) 0.3

Solution

a) $0.0015 = \frac{15}{10,000}$

$$= \frac{1.5 \times 10}{10,000}$$

$$= \frac{1.5}{10^3}$$

$$= 1.5 \times 10^{-3}$$

(Since from the 4th law of indices $1/x^a = x^{-a}$)

b) $0.000026 = \frac{26}{1,000,000}$

$$= \frac{2.6 \times 10}{1,000,000}$$

$$= \frac{2.6}{100,000}$$

$$= \frac{2.6}{10^5}$$

$$= 2.6 \times 10^{-5}$$

$$\begin{aligned} \text{c) } 0.000000067 &= \frac{67}{1,000,000,000} \\ &= \frac{6.7 \times 10}{1,000,000,000} \\ &= \frac{6.7 \times 1}{1,000,000,000} \\ &= 6.7 \times \frac{1}{10^8} \\ &= 6.7 \times 10^{-8} \end{aligned}$$

$$\begin{aligned} \text{d) } 0.3 &= \frac{3}{10} \\ &= \frac{3 \times 1}{10^1} \\ &= 3 \times 10^{-1} \end{aligned}$$

4. Express the following as decimal fractions

(a) 9.4×10^{-5} (b) 8.8×10^{-3} (c) 1.8×10^{-1} (d) 2×10^{-7}

Solutions

a) 9.4×10^{-5}

$$\frac{9.4 \times 1}{10^5}$$

(by using the 4th law of indices as explained in the example 3 above)

$$\begin{aligned} &= \frac{9.4}{100,000} \\ &= 0.000094 \end{aligned}$$

b) $8.8 \times 10^{-3} = \frac{8.8 \times 1}{10^3}$

$$\begin{aligned} &= \frac{8.8}{1000} \\ &= 0.0088 \end{aligned}$$

c) $1.8 \times 10^{-1} = \frac{1.8 \times 1}{10^1}$

$$\begin{aligned} &= \frac{1.8}{10} \\ &= 0.18 \end{aligned}$$

d) $2 \times 10^{-7} = \frac{2 \times 1}{10^7}$

$$\begin{aligned} &= \frac{2 \times 1}{10,000,000} \\ &= 0.0000002 \end{aligned}$$

Note that for decimal fraction, n is a negative integer

EVALUATION

1. Change (a) 9.18×10^5 (b) 6.75×10^{-8} to ordinary number
2. Express the following in standard form (a) 0.0000058 (b) 458000

5.2 ROUNDING OFF NUMBERS

When rounding off number digits 1,2,3,4 are rounded down and digits 5, 6, 7, 8, 9 are rounded up.

Examples:

Round off the following to the nearest

- a) Thousand
- b) Hundred
- c) Ten

- 1. 4517
- 2. 30,637

Solutions

- a) $4517 \approx 500$ to the nearest thousand
- b) $4517 \approx 4500$ to the nearest hundred
- c) $4517 \approx 4520$ to the nearest tens
- 2. (a) $30,637 \approx 31,000$ to the nearest thousand
- (b) $30,637 \approx 30,600$ to the nearest hundred
- (c) $30,637 \approx 30,640$ to the nearest ten

SIGNIFICANT FIGURES

The significant figures begin from the first non-zero digit at the left hand side of a number. As before, digits 1,2,3,4, are rounded down and digits 5,6,7,8,9 are rounded up. Digits should be written with their correct place value.

Note that zero in between non-zero digits in a number are significant. E.g the zero in 8.0296 is a significant while zero in 0.0000925 are not significant.

Examples:

Round off the following to;

- a) 1 significant figure.
- b) 2 significant figures.
- c) 3 significant figures.

- 1. 26,002
- 2. 2.00567
- 3. 0.006307

Solution

- 1. (a) $26,002 \approx 30,000$ to 1 significant figure.
- (b) $26,002 \approx 26,000$ to 2 significant figures.
- (c) $26,002 \approx 26,000$ to 3 significant figures.
- 2. (a) $2.00567 \approx 2$ to 1 significant figure.
- (b) $2.00567 \approx 2.0$ to 2 significant figures.
- (c) $2.00567 \approx 2.01$ to 3 significant figures.
- 3. (a) $0.006307 \approx 0.006$ to 1 significant figure.
- (b) $0.006307 \approx 0.0063$ to 2 significant figures.
- (c) $0.006307 \approx 0.00631$ to 3 significant figures.

5.3 DECIMAL PLACES

Decimal places are counted from the decimal point. Zero after the point is significant and also counted. Digits are rounded up and down as before. Place value must be kept.

Examples:

Round off the following to:

- (a) 1 d.p (b) 2 d.p (c) 3 d.p

1. 0.0089
2. 0.9002
3. 1.9875

Solutions

1. (a) $0.0089 \approx 0.0$ to 1 decimal place.
(b) $0.0089 \approx 0.01$ to 2 decimal places.
(c) $0.089 \approx 0.009$ to 3 decimal places.
2. (a) $0.9002 \approx 0.9$ to 1 decimal place.
(b) $0.9002 \approx 0.90$ to 2 decimal places.
(c) $0.9002 \approx 0.900$ to 3 decimal places.
3. (a) $1.9875 \approx 2.0$ to 1 decimal place.
(b) $1.9875 \approx 1.99$ to 2 decimal places.
(c) $1.9875 \approx 1.988$ to 3 decimal places.

EVALUATION

1. Express the following in standard form
(a) 3,500,000 (b) 28 (c) 0.47 (d) 0.0000003
2. In the following statements round each number to two significant figures then write it out in full:
(a) It will cost #3.28 billion to renovate state's classrooms (b) The area of Ghana is 23.9 million hectares
(b) It was estimated that the population of Lagos was about 9.44 millions in 2000

READING ASSIGNMENT

NGM SSS1, pages 6-7, review test 3 and 4.

GENERAL EVALUATION

1. Express the following in standard form (a) 0.000423 (b) 628500
2. (a) Change 4.23×10^7 to ordinary form (b) Change 3.4×10^{-6} to decimal fraction
3. Approximate 72899 to the nearest (a) Ten (b) hundred (c) thousand
4. approximate 0.0065734 to (a) 1 s.f (b) 2 s.f (c) 3 s.f
5. Approximate 99.99054 to (a) 1 d.p (b) 2 d.p (c) 3 d.p

WEEKEND ASSIGNMENT

1. Round off 0.004365 correct to 2 significant figures (a) 0.04 (b) 0.0044 (c) 0.00437 (d) 0.0043 (e) 0.44
2. What is 0.002568 correct to 3 decimal places (a) 0.00 (b) 0.002 (c) 0.003 (d) 0.00256 (e) 0.00257
3. Write 0.0000549 in standard form (a) 5.49×10^{-5} (b) 5.49×10^{-4} (c) 5.49×10^{-3} (d) 5.49×10^4 (e) 5.49×10^5
4. Write 5.48×10^{-4} as decimal fraction (a) 0.0548 (b) 0.00548 (c) 0.000548 (d) 0.0000548 (e) 0.00000548
5. If 0.00725 is written as 7.25×10^n , the value of n is _____ (a) -4 (b) -3 (c) -2 (d) -1 (e) 3

THEORY

1. The Page of a book are numbered 1 to 300
(a) How many thicknesses of the paper make 300 pages
(b) If the thickness of the book is 15mm, calculate the thickness of one leaf. Give your answer in meters in standard form
2. A length of wire is given as 6.8cm correct to 2 significant figures. What is the least possible length of the wire
(a) Give the number 29,542 to the nearest ten (b) Write 0.07258 to 3 significant figures

STUDY SESSION 6: INDICES

INTRODUCTION

This session provides readers with a detailed explanation of Indices, a fundamental concept in mathematics used to represent repeated multiplication in a concise and systematic form. It introduces the concept of indices and explains the relationship between bases, powers, and exponents.

The session explains the basic laws and principles of indices, including the product law, quotient law, power of a power, zero index, negative indices, and fractional indices. It also demonstrates how these laws can be applied to simplify mathematical expressions involving powers.

In addition, the session covers the procedures for evaluating expressions involving indices and simplifying expressions using the appropriate laws. Suitable examples are provided to improve understanding and accuracy when working with numbers and algebraic expressions.

The session also highlights the importance of indices in mathematics, science, engineering, and computer science. The systematic application of the laws of indices provides learners with essential skills for solving mathematical problems involving powers, roots, and exponential expressions.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define indices and explain the meaning of a base and an exponent.
- ii. Explain the relationship between powers and repeated multiplication.
- iii. Identify and apply the laws of indices to simplify mathematical expressions.
- iv. Apply the product and quotient laws of indices to appropriate mathematical expressions.
- v. Apply the power of a power law when simplifying expressions involving indices.
- vi. Explain and use the laws of zero and negative indices correctly.
- vii. Evaluate expressions involving fractional indices and relate them to roots.
- viii. Simplify numerical and algebraic expressions using the appropriate laws of indices.
- ix. Solve basic mathematical problems involving indices using systematic procedures.
- x. Apply knowledge of indices to more advanced mathematical concepts involving powers, roots, and exponential expressions.

6.1 LAWS OF INDICES

1st to 4th laws for all values of a , b and $x \neq 0$

1. $X^a \times X^b = X^{a+b}$
2. $X^a \div X^b = X^{a-b}$
3. $X^0 = 1$

$$4. X^{-a} = \frac{1}{X^a}$$

Examples:

Simplify

$$1. 10^5 \times 10^4 \quad 2. a^3 \times a^4 \quad 3. m^8 \div m^5 \quad 4. 24x^6 \div 8x^4 \quad 5. 19^8 \div 19^8$$

Solutions

$$1. 10^5 \times 10^4 = 10^{5+4} = 10^9$$

$$2. a^3 \times a^4 = a^{3+4} = a^7$$

$$3. m^8 \div m^5 = m^{8-5} = m^3$$

$$4. 24x^6 \div 8x^4 = \frac{24x^6}{8x^4} = 3x^{6-4} = 3x^2$$

$$5. 19^8 \div 19^8 = 19^{8-8} = 19^0 = 1$$

Evaluation

Simplify

$$a) 6 \times 7^0 \quad (b) 4^{-3} \quad (c) Z^3 \times (\frac{1}{6})^1 \quad (d) r \times r \times r \times r \times r^{-5}$$

6.2 PRODUCT OF INDICES

$$(X^a)^b = X^{a \times b} = X^{ab}$$

Examples

Simplify

$$1. (X^2)^3 \quad 2. (Y^4)^2 \quad 3. (3^{-2})^{-3} \quad 4. (-3d^3)^2 \quad 5. a^6(-a)^{-4}$$

Solutions

$$1. (X^2)^3 = X^{2 \times 3} = X^6$$

$$2. (Y^4)^2 = Y^{4 \times 2} = Y^8$$

$$3. (3^{-2})^{-3} = 3^{-2 \times -3} = 3^{+6} \\ = 3^6 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \\ = 27 \times 27 \\ = 729$$

$$4. (-3d^3)^2 = (-3)^2 \times (d^3)^2 \\ = -3 \times -3d^6 = 9d^6$$

$$5. a^6(-a)^{-4} = \frac{a^6 \times 1}{(-a)^4} \\ = \frac{a^6}{(-a) \times (-a) \times (-a) \times (-a)} \\ = a^6 \frac{1}{a^4} \\ = a^{6-4} \\ = a^2$$

EVALUATION

Simplify

$$1. (h^4)^{-5} \quad 2. (-4u^2v)^3 \quad 3. (-x^3)^2 \div x^4 \quad 4. -(d^2) \div d^4 \times -d \quad 5. (-c)^2 \times (c)^4 \div (-c^3)$$

6.3 FRACTIONAL INDICES

$$X^{1/a} \text{ and } X^{a/b}$$

$\sqrt{X}$ is short for the square root of x

$$\sqrt{X} \times \sqrt{X} = X$$

$$\text{Let } \sqrt{x} = x^p$$

Then

$$X^p \times X^p = \sqrt{x} \times \sqrt{x} = x^1$$

By equating the indices

$$2p = 1, \quad p = \frac{1}{2}$$

$$\text{Thus } \sqrt{x} = x^{1/2}$$

Similarly, $\sqrt[3]{x}$ is the short for the cube root of x e.g. $\sqrt[3]{8} = 2$. Since $2 \times 2 \times 2 = 8$

$$\text{And } \sqrt[3]{-27} = -3$$

$$\text{Since } (-3) \times (-3) \times (-3) = -27$$

$$\sqrt[3]{x} \times \sqrt[3]{x} \times \sqrt[3]{x} = x$$

$$\text{i.e. } x^q \times x^q \times x^q = x^1$$

$$x^{3q} = 1$$

Equating the power

$$3q = 1$$

$$q = \frac{1}{3}$$

$$\text{thus } \sqrt[3]{x} = x^{1/3}$$

$$\text{In general } x^{1/a} = \sqrt[a]{x}$$

$$\text{Also } x^{2/3} = x^2 \times \frac{1}{3} = (x^2)^{1/3}$$

$$= \sqrt[3]{x^2}$$

OR

$$x^{2/3} = (x^2 \times \frac{1}{3}) = (x^{1/3})^2$$

$$= (\sqrt[3]{x})^2$$

In general

$$x^{a/b} = \sqrt[b]{x^a} \text{ or } (\sqrt[b]{x})^a$$

Examples

Simplify

$$1. \ 8^{-2/3} \quad 2. \ 4^{1/6} \times 4^{1/3} \quad 3. \ (16/81)^{-3/4}$$

$$4. \sqrt{72a^3b^{-2}/2b^5b^{-6}}$$

Solution

$$1. \ 8^{-2/3} = \frac{1}{8^{2/3}}$$

$$= \frac{1}{(\sqrt[3]{8})^2}$$

$$= \frac{1}{(2)^2} = \frac{1}{4}$$

$$2. \ 4^{1/6} \times 4^{1/3} = 4^{1/6 \div 1/3}$$

$$= 4^{3/6} = 4^{1/2}$$

$$= \sqrt{4} = 2$$

$$3. \ (16/81)^{-3/4} = \frac{1}{(16/81)^{3/4}}$$

$$= \frac{1}{(\sqrt[4]{16/81})^3}$$

$$= \frac{1}{(2/3)^3}$$

$$= 1 \div (2/3)^3$$

$$= 1 \div 8/27 = 1 \times 27/8 = 27/8$$

$$\begin{aligned}
 4. \sqrt{\frac{72a^2b^{-2}}{2b^5b^{-6}}} &= \sqrt{\frac{(72a^3b^{-2})^{1/2}}{2a^5b^{-6}}} \\
 &= \sqrt{\frac{72 \times a^3 \times a^{-5} \times b^2 \times b^{-6}}{2}} \\
 &= \sqrt{36a^{-2} \times b^{-4}} \\
 &= \sqrt{36a^{-2} \times b^{-2+6}} \\
 &= \sqrt{36a^{-2} \times b^4} \\
 &= \sqrt{36 \times (a^{-2} \times b^4)^{1/2}} \\
 &= 6 \times a^{-2 \times 1/2} \times b^{4 \times 1/2} \\
 &= 6a^{-1} \times b^2 = 6 \times \frac{1}{a} \times b^2 \\
 &= \frac{6b^2}{a}
 \end{aligned}$$

EVALUATION

Simplify 1. $(125)^{-1/3}$ 2. $(18/32)^{-3/2}$ 3. $(\sqrt[3]{4})^{1.5}$ 4. $64^{-5/6}$ 5. $\sqrt[3]{19/16}$

6.4 SOLVING EQUATION WITH INDICES

Solve the following equations:

- a) $2r^{-3} = -16$
 b) $\frac{5x}{5} = \frac{40}{5} x^{-1/2}$
 c) $4^{c-1} = 64$

Solutions

1. $2r^{-3} = -16$

Divide both sides by 2

$$\frac{2r^{-3}}{2} = \frac{-16}{2}$$

$$r^{-3} = -8$$

$$\frac{1}{r^3} = \frac{-8}{1}$$

$$-8r^3 = 1 \times 1$$

$$r^3 = \frac{-1}{8}$$

Take cube root of both sides

$$\sqrt[3]{r^3} = \sqrt[3]{\frac{-1}{8}}$$

$$r = \frac{-1}{2}$$

2. $\frac{5x}{5} = \frac{40x^{-1/2}}{5}$

$$x = 8x^{-1/2}$$

$$x = 8 \times \frac{1}{x^{1/2}}$$

$$x^{1/2}$$

Cross multiply

$$x \times x^{1/2} = 8$$

$$x^1 \times x^{1/2} = 8$$

$$x^{1+1/2} = 8$$

$$x^{3/2} = 8$$

$$\text{i.e. } (\sqrt{x})^3 = 8$$

raise both sides by power $2/3$

$$(x^{3/2})^{2/3} = (8)^{2/3}$$

$$x^1 = (\sqrt[3]{8})^2$$

$$x = (2)^2$$

$$x = 4$$

3. $4^{c-1} = 64$

Change both sides to the same base

$$4^{c-1} = 4^3$$

Equate the powers

$$c-1 = 3$$

$$c = 3 + 1$$

$$c = 4$$

EVALUATION

Solve the following equation

1. $a^{2/3} = 9$ 2. $2x^3 = 54$

GENERAL EVALUATION

1. If $9^{2x+1} = \frac{81^{x-2}}{3^x}$ find x

2. Solve $9^{x-1} = 27^{x+1}$

3. Simplify $\sqrt[3]{\frac{72p^{-3}q^{-7}}{9p^9q^5}}$

READING ASSIGNMENT

NGM SSS page 18, exercise 1d numbers 21-50.

WEEKEND ASSIGNMENT

- Simplify $3^3 \times 6^{-3} \times 2^5$ (a) 0 (b) 1 (c) 2 (d) 4 (e) 12
- Calculate the value of $(27/125)^{1/3} \times (4/9)^{1/2}$ (a) 12/25 (b) 2/5 (c) 3/5 (d) 9/10 (e) 10/9
- If $5p^{-3} = 8 \times 5^{-2}$, find the value of p (a) 8/125 (b) 2/5 (c) 4/5 (d) 8/5 (e) 5/2
- If $x^2 = 81^{1/2}$, $x =$ ----- (a) 3 (b) 9 (c) 18 (d) 27 (e) 54
- Simplify $(x^{1/2})^3 \times \frac{1}{x^4}$ (a) 1 (b) x^1 (c) x^3 (d) x^0 (e) 2

THEORY

- Evaluate $\frac{9^{1/2} \times 27^{2/3}}{64^{1/3}}$
- Solve the following equations
 - $Y^{-2} = 9$
 - $(2s)^{1/2} = 9$
 - $2^{n-1} = 16$

STUDY SESSION 7: LOGARITHMS

7.0 INTRODUCTION

This session provides readers with a detailed explanation of Logarithms, a fundamental concept in mathematics that is closely related to indices and exponents. It introduces the concept of logarithms and explains how they can be used to determine the power to which a given base must be raised to obtain a particular number.

The session explains the basic principles and laws of logarithms, including the product law, quotient law, and power law. It also discusses common logarithms, logarithms to other bases, and the relationship between logarithmic and exponential forms.

In addition, the session covers the procedures for evaluating, simplifying, and solving logarithmic expressions and equations. Suitable examples are provided to demonstrate the application of logarithmic laws and to improve understanding and accuracy.

The session also highlights the importance of logarithms in mathematics, science, engineering, computing, and other practical applications. The systematic methods for working with logarithms are presented to develop students' problem-solving skills and prepare them for more advanced mathematical concepts.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define logarithms and explain their relationship with indices and exponents.
- ii. Identify the base and argument of a logarithm and interpret logarithmic notation correctly.
- iii. Convert between logarithmic and exponential forms of mathematical expressions.
- iv. Evaluate simple logarithmic expressions using appropriate methods.
- v. State and apply the laws of logarithms, including the product, quotient, and power laws.
- vi. Simplify logarithmic expressions using the appropriate laws and properties.
- vii. Solve simple logarithmic equations using systematic procedures.
- viii. Explain the relationship between logarithms and exponential functions.
- ix. Apply logarithmic techniques to solve basic mathematical and real-life problems.
- x. Recognize the applications of logarithms in areas such as science, engineering, computing, finance, and data analysis.

7.1 LOGARITHMS OF NUMBERS TO BASE 10

In general the logarithm of a number is the power to which the base must be raised in order to give that number. i.e if $y=n^x$, then $x = \log_n y$. Thus, logarithms of a number to base ten is the power to which 10 is raised in order to give that number i.e if $y = 10^x$, then $x = \log_{10} y$. With this definition $\log_{10} 1000 = 3$ since $10^3 = 1000$ and $\log_{10} 100 = 2$ since $10^2 = 100$.

Examples:

1. Express the following in logarithms form

a) $2^{-6} = 1/64$ b) $3^5 = 243$ c) $5^3 = 125$ d) $10^4 = 10,000$

Solutions

1. (a) $2^{-6} = \frac{1}{64}$
 $= \log_2 (1/64) = -6$

(b) $3^5 = 243$
 $= \log_3 243 = 5$

(c) $5^3 = 125$
 $= \log_5 125 = 3$

(d) $10^4 = 10,000$
 $= \log_{10} 10000 = 4$

2. Express the following in index form

a) $\log_2(1/8) = -3$ (b) $\log_{10}(1/100) = -2$ (c) $\log_4 64 = 3$ (d) $\log_5 625 = 4$ (e) $\log_{10} 1000 = 3$

Solutions

a) $\log_2 (1/8) = -3$
Then $2^{-3} = 1/8$

b) $\log_{10}(1/100) = -2$
Then $10^{-2} = 1/100$

c) $\log_4 64 = 3$
Then $4^3 = 64$

d) $\log_5 625 = 4$
Then $5^4 = 625$

e) $\log_{10} 1000 = 3$
Then $10^3 = 1000$

Note: Logarithms of numbers to base ten are found with the help of tables

Examples:

Use the tables to find the log of:

(a) 37 (b) 3900 to base ten

Solutions

1. $37 = 3.7 \times 10$
 $= 3.7 \times 10^1$ (standard form)
 $= 10^{0.5682 + 1} \times 10^1$ (from table)
 $= 10^{1.5682}$
Hence $\log_{10} 37 = 1.5682$

$$\begin{aligned}
 2. \quad 3900 &= 3.9 \times 1000 \\
 &= 3.9 \times 10^3 \text{ (standard form)} \\
 &= 10^{0.5911} \times 10^3 \text{ (from table)} \\
 &= 10^{0.5911 + 3} \\
 &= 10^{3.5911}
 \end{aligned}$$

$$\text{Therefore } \log_{10} 3900 = 3.5911$$

In logarithms any of the number there are two parts, an integer (whole number) before the decimal point and a fraction after the decimal point which is also called mantissa. E.g

$$\log_{10} 3900 = 3.5911$$

Integer

decimal fraction (mantissa)

The integer part of $\log_{10} 3900$ is 3 and the decimal part is .5911

In order to obtain the integer part of the logarithm of a number to base ten, count the number of digits to the left hand side of the decimal point and subtract 1. The decimal fraction part of the logarithm of the given number is obtained from the tables.

Examples:

Use the logarithm table to find the logarithms to base ten of:

1. 51.38 2. 840.3 3. 65160

Solutions

1. $\log_{10} 51.38 = 1.7108$
2. $\log_{10} 840.3 = 2.9244$
3. $\log_{10} 65160 = 4.8140$

7.2 Antilogarithms table

Antilogarithm is the opposite of logarithms. To find number whose logarithm is given. It is possible to use logarithm table in reverse

However, it's convenient to use the tables of antilogarithms. When finding an antilogarithm, look up the fractional part only, then used the integer to place correct the decimal point correctly in the final number

Examples:

Find the antilog of the following logarithms:

- a) 0.5682
- b) 2.7547
- c) 5.3914

Solutions

Log	antilog
a) 5682	3.700
b) 2.7547	568.4
c) 5.3914	246200

Logarithms of numbers less than 1

No	Log	Antilog
8320	3.9201	8320
58.24	1.7652	58.24

Evaluation

1. Find the log of: (i) 0.009321 (ii) 0.5454
- (b) Find the antilog of: (iii) 3.3210 (iv) 1.8113 (v) 0.5813 (vi) 3.2212

7.3 MULTIPLICATION AND DIVISION OF NUMBERS USING LOGARITHMS TABLES

Evaluate the following using tables

1. 4627×29.3
2. $819.8 \div 3.905$
3. $\frac{48.63 \times 8.53}{15.39 \times 3.52}$

Solutions

1. 4627×29.3

No	Log
4627	3.6653
29.3	+ 1.4669
135600	5.1322

$$4627 \times 29.3 = 135600 \text{ (4 s.f)}$$

2. $819.8 \div 3.905$

No	Log
819.8	2.9137
3.905	- 0.5916
209.9	2.3221

$$\text{Therefore } 819.8 \div 3.905 = 209.9$$

3. $\frac{48.63 \times 8.53}{15.39 \times 3.52}$

No	log
48.63	+ 1.6869
8.53	+ 0.9309
Numerator 2.6178	2.6178
15.39	+ 1.1872
3.52	+ 0.5465
Denominator 1.7337	1.7337
7.658	0.8841

$$\text{Therefore } \frac{48.63 \times 8.53}{15.39 \times 3.52} = 7.658$$

$$15.39 \times 3.52$$

EVALUATION

Use logarithms tables to calculate

1. $\frac{36.12 \times 750.9}{113.2 \times 9.98}$ (2) 3577×31.11 (3) $256.5 \div 6.45$

GENERAL EVALUATION

1. Change the following to logarithms form
 - a. $25^{1/2} = 5$
 - b. $(0.01)^2 = 0.0001$
2. Change the following to index form
 - a. $\log_{31} 1 = 0$
 - b. $\log_{15} 225 = 2$
3. Evaluate the following using logarithms tables
 - a. $\frac{69.7 \times 44.63}{25.67}$
 - b. $17.9 \times 3.576 \times 98.14$

READING ASSIGNMENT

New General mathematics SSS1, page 21, Exercise 1h 1 – 3.

WEEKEND ASSIGNMENT

- Find the log of 802 to base 10 (use log tables) (a) 2.9042 (b) 3.9040 (c) 8.020 (d) 1.9042
- Find the number whose logarithm is 2.8321 (a) 6719.2 (b) 679.4 (c) 0.4620 (d) 67.92
- What is the integer of the log of 0.000352 (a) 4 (b) $\overline{3}$ (c) $\overline{4}$ (d) 3
- Given that $\log_2(1/64) = m$, what is m ? (a) -5 (b) -4 (c) -6 (d) 3
- Express the log in index form: $\log_{10} 10000 = 4$ (a) $10^3 = 10000$ (b) $10^{-4} = 10000$ (c) $10^4 = 10000$ (d) $10^5 = 100000$

THEORY

- Evaluate using logarithm table $\frac{6.28 \times 304}{981}$

And express your answer in the form $A \times 10^n$, where A is a number between 1 and 10 and n is an integer.

- Use logarithm table to calculate $\frac{6354 \times 6.243}{16.76 \times 323}$ correct to 3 s.f

7.5 POWERS

Number	Power of 10
1000	10^3
100	10^2
10	10^1
1	0
0.1	10^{-1}
0.01	10^{-2}
0.001	10^{-3}

The table above shows that 1000 is to the power of 3, thus the logarithm of 1000 to base 10 is 3. In this case, the logarithm of a number is the power to which 10 is raised to give that number. Thus a logarithm is another word for power or index.

Logarithms can be found in other bases apart from base 10

For example, since $32 = 2^5$, then $\log_2 32 = 5$ i.e log of 32 to base 2 is 5. In general, if $y = n^x$, then $x = \log_n y$

For logarithms to base 10, the following table can be stated:

Power (indices)	logarithms (indices)
$1000 = 10^3$	$\log 1000 = 3$
$100 = 10^2$	$\log 100 = 2$
$10 = 10^1$	$\log 10 = 1$
$1 = 10^0$	$\log 1 = 0$

Thus, a statement written in index form can be changed to a logarithm form and vice versa.

Examples:

- Express the following in logarithms form

- $2^{-3} = 1/8$
- $3^6 = 729$
- $4^3 = 256$

Solutions

- $2^{-3} = 1/8$

- Then, $\log_2 1/8 = 3$
- b. $3^6 = 729$
Then, $\log_3 729 = 6$
- c. $4^3 = 256$
Then, $\log_4 256 = 3$
2. Express the following in index form
- a. $\log_{10} \frac{1}{100} = 3$
- b. $\log_2 64 = 6$
- c. $\log_5 (1/125) = -3$

Solution

1. $\log_{10} \frac{1}{1000} = -3$
Then $10^{-3} = 1/1000$
2. $\log_2 64 = 6$
Then $2^6 = 64$
3. $\log_5 (1/125) = -3$
Then $5^{-3} = 1/125$

EVALUATION

1. Given that $\log_3 81 = m$, then $3^m = 81$. What is m ?
2. Find the value of $\log_2 128$
3. Fill in the blank box in the statement below
 $\log_{\square} 343 = 3$

7.6 CALCULATION OF POWERS AND ROOTS USING LOGARITHM TABLES

When solving problems of powers and roots using logarithms tables, first find the logarithm of the number and then apply the multiplication or division law of indices to the logarithm value i.e multiply the power with the logarithm and divide the logarithm by root

Examples

Evaluate using logarithm table

1. 252.8^2 2. $\sqrt[6]{35.81}$ 3. $\sqrt{26.21}$

Solutions

1. 252.8^2
- | No | log |
|----------------|-------------------|
| 252.8^2 | 2.4028×2 |
| Antilog= 63920 | $= 4.8056$ |
- Therefore $252.8^2 = 63920 = 63900$

2. $\sqrt[6]{35.81}$
- | No | log |
|-------------------|-----------------|
| $\sqrt[6]{35.81}$ | $1.5540 \div 6$ |
| Antilog= 1.816 | $= 0.2590$ |
- Therefore $\sqrt[6]{35.81} = 1.816$

3. $\sqrt{26.21}$
- | No | log |
|----------------|-----------------|
| $\sqrt{26.21}$ | $1.4185 \div 2$ |
| 5.121 | 0.7027 |
- Therefore $\sqrt{26.21} = 5.121$

EVALUATION

Evaluate using logarithms tables

1. 3.53^3 2. $\sqrt[4]{400}$

7.7 CALCULATION INVOLVING MULTIPLE DIVISION, POWERS AND ROOTS USING LOGARITHMS

Example

Evaluate the following using logarithms tables correct to 3.s.f

1. $\frac{94100 \times 38.2}{5.8^3 \times 8.14}$
 2. $319.63 \times 12.28^2 \times 74$
 3. $\sqrt[3]{\frac{218 \times 37.2}{95.43}}$
 4. $\sqrt[3]{\frac{38.32 \times 38.2}{8.637 \times 6.285}}^2$

Solution

1. $\sqrt[3]{94100 \times 38.2}$

No	log		
$\sqrt[3]{94100}$	$4.9736 \div 2$		2.4868
38.2	1.5821		+1.5821
Numerator		4.0689	
5.69^3	0.754×3		2.2629
8.14	0.9106		+0.9106
denominator		3.1735	
numerator	4.0689		
denominator	- 3.1735		
7.859		0.8954	

Therefore $\frac{94100 \times 38.2}{5.582 \times 8.14} = 7.859$

2. $\sqrt[3]{19.63 \times 12.28^2 \times 74}$

No	log		
19.63	1.2930		1.2930
12.28^2	1.089×2	+	2.1784
74	1.8692		1.8692
$\sqrt[3]{19.63 \times 12.28^2 \times 74}$			$5.3406 \div 3$
60.29			1.7802

Therefore $\sqrt[3]{19.63 \times 12.28^2 \times 74} = 60.29 = 60.3$ to 3.s.f

3. $\sqrt[3]{\frac{19.63 \times 37.2}{95.43}}$

No	log		
218	2.338		
37.2	+	1.5705	
Numerator	-	3.9090	
Deno. 95.43		1.9797	

4.397

$$\begin{array}{r} 1.9294 \div 3 \\ 0.6431 \end{array}$$

$$\sqrt[3]{\frac{19.63 \times 37.2}{95.43}} = 4.397 = 4.40 \text{ to 3 s.f}$$

$$4. \sqrt[3]{\frac{38.32 \times 2.961^2}{8.637 \times 6.285}}$$

No	log	
38.32	1.5834	
2.961	0.4719	
Numerator	2.0553	2.0553
8.637	0.9364	
6.285	0.7983	
Denominator	1.7347	1.7347
	0.3206 X 2	
	0.6412 ÷ 3	
1.636	0.2137	

$$\sqrt[3]{\frac{38.32 \times 2.961^2}{8.637 \times 6.285}} = 1.636$$

EVALUATION

Calculate the following

$$1. \sqrt[3]{\frac{1064}{29.4}}$$

$$2. \sqrt[3]{\frac{403.9 \times 5.78^2}{70.62 \times 2.931}}$$

GENERAL EVALUATION

1. If $\log 0.04 = m$ and $5m = 0.04$ find the value of m

2. Evaluate the following using logarithms table

a) $(35.61)^2 \times 5.62$

$$\sqrt[3]{143.5}$$

b) $\sqrt[3]{\frac{634.9^2}{21.5}}$

READING ASSIGNMENT

NGM SSS1, pages 21-22, exercise 1h no 4 - 12.

WEEKEND ASSIGNMENT

1. Use the table to find the logarithm of 3.7 (a) 0.5682 (b) 1.5682 (c) 2.5682 (d) 3.5682

2. Evaluate $\sqrt[3]{35}$ (a) 7.047 (b) 7047 (c) 704.7 (d) 0.7047

3. Write down the integer of $\log 25.82$ (a) 0 (b) 1 (c) 2 (d) 1
4. Use table to find the log of 12.34 (a) 12.35 (b) 1.09913 (c) 2.0913 (d) 1.234
5. Find the number whose logarithms is 2.8321 (a) 679.2 (b) 679.4 (c) 0.4620 (d) 46.2

THEORY

1. Use the logarithms tables to evaluate

$$\frac{28.61^2 \times 74.23}{355.6 \times 2.547}$$

2. $\frac{403.9^3}{79.62}$

STUDY SESSION 8: SIMPLE EQUATION AND VARIATIONS

INTRODUCTION

This session provides readers with a detailed explanation of **Simple Equations and Variations**, fundamental concepts in mathematics used to represent relationships between quantities and solve unknown values. It introduces the concept of simple equations and explains how mathematical statements can be formed and solved using algebraic operations.

The session explains the principles and procedures involved in solving **simple linear equations**, including equations involving addition, subtraction, multiplication, and division. It also demonstrates how algebraic expressions can be simplified and how unknown quantities can be determined systematically.

In addition, the session introduces the concept of **variation**, which describes the relationship between two or more quantities. It covers **direct variation, inverse variation, and joint variation**, with suitable examples to demonstrate how these relationships can be represented and solved mathematically.

The session also highlights the importance of simple equations and variations in mathematics, science, economics, engineering, and everyday problem-solving. The systematic procedures for solving equations and variation problems are presented to improve logical reasoning, accuracy, and mathematical problem-solving skills.

LEARNING OUTCOMES FOR THE STUDY SESSION

When you have studied this session, you should be able to:

- i. Define a simple equation and explain the meaning of an unknown variable.
- ii. Identify the different components of a simple equation, including variables, constants, coefficients, and operators.
- iii. Solve simple linear equations involving addition, subtraction, multiplication, and division.
- iv. Apply appropriate algebraic operations to simplify and solve equations.
- v. Translate simple word problems into mathematical equations and solve them.
- vi. Define variation and explain the relationship between two or more varying quantities.
- vii. Explain and distinguish between direct, inverse, and joint variation.
- viii. Solve problems involving direct and inverse variations using appropriate methods.
- ix. Form variation equations from given conditions and determine unknown constants or quantities.
- x. Apply simple equations and variations to solve basic mathematical and real-life problems.

8.1 EQUATIONS

An equation is a statement of two algebraic expressions which are equal in value. For example, $4 - 4x = 9 - 12x$ is a linear equation with an unknown x . This equation is only true when x has a particular numerical value. To solve an equation means to find the real number value of the unknown that makes the equation true.

Example 1

Solve $4 - 4x = 9 - 12x$

$$4 - 4x = 9 - 12x$$

Add $12x$ to both sides of the equation.

$$4 - 4x + 12x = 9 - 12x + 12x$$

$$4 + 8x = 9$$

Subtract 4 from both sides of the equation.

$$4 + 8x - 4 = 9 - 4$$

$$8x = 5$$

Divide both sides of the equation by 8

$$\frac{8x}{8} = \frac{5}{8}$$

$$x = \frac{5}{8}$$

$\frac{5}{8}$ is the solution or root of the equation.

Check: when $x = \frac{5}{8}$,

$$\text{LHS} = 4 - 4 \times \frac{5}{8} = 4 - 2\frac{1}{2} = 1\frac{1}{2}$$

$$\text{RHS} = 9 - 12 \times \frac{5}{8} = 9 - 7\frac{1}{2} = 1\frac{1}{2} = \text{LHS}$$

The equation in Example 1 was solved by the balance method. Compare the equation with a pair of scales. If the expressions on opposite sides of the equals sign 'balance', they will continue to do so if the same amounts are added to or subtracted from both sides. They will also balance if both sides are multiplied or divided by the same amounts.

Example 2

Solve $3(4c - 7) - 4(4c - 1) = 0$

Remove brackets.

$$12c - 21 - 16c + 4 = 0$$

Collect like terms.

$$-4c - 17 = 0$$

Add 17 to both sides

$$-4c = 17$$

Divide both sides by -4.

$$c = -\frac{17}{4}$$

$$c = -4\frac{1}{4}$$

Check: When $c = -4\frac{1}{4}$,

$$\text{LHS} = 3(-17 - 7) - 4(-17 - 1)$$

$$= 3(-24) - 4(-18)$$

$$-72 + 72 = 0 = \text{RHS}$$

EVALUATION

Solve the following equations

1. $2 - 5t = 20 - 8t$

2. $d = 12 - (11 + 4d)$

3. $2\frac{1}{3}d = 28$

8.2 Change of Subject of Formulae

It is often necessary to change the subject of a formula. To do this, think of the formula as an equation. Solve the equation for the letter which is to become the subject. The following examples show how various formulae can be rearranged to change the subject.

Example 1

Make x the subject of the formula $a = b(a - x)$

Clear brackets.

$$a = b - bx$$

rearrange to give terms in x on one side of the equation.

$$bx = b - a$$

divide both sides by b .

$$x = \frac{b - a}{b}$$

Example 2

make x the subject of the formula $a = \frac{b + x}{b - x}$

$$a = \frac{b + x}{b - x}$$

clear fractions. Multiply both sides by $(b - x)$

$$ab - ax = b + x$$

collect terms in x on one side of the equation.

$$ab - b = ax + x$$

take x outside a bracket (factorise).

$$ab - b = x(a + 1)$$

divide both sides by $(a + 1)$.

$$\frac{ab - b}{a + 1} = x$$

$$\therefore x = \frac{ab - b}{a + 1} = \frac{b(a - 1)}{a + 1}$$

Example 3

Make x the subject of the formula

$$b = \frac{1}{2}\sqrt{a^2 - x^2}$$

clear fractions.

$$2b = \sqrt{a^2 - x^2}$$

Square both sides.

$$(2b)^2 = a^2 - x^2$$

$$4b^2 = a^2 - x^2$$

Rearrange to give the term in x on one side of the equation.

$$x^2 = a^2 - 4b^2$$

Take the square root of the both sides.

$$x = \pm \sqrt{a^2 - 4b^2}$$

The general method of Example 1, 2 and 3 is to treat the formula as an equation and the new subject as the unknown of the equation.

There are many different formulae. Therefore it is not possible to give general rules for changing their subject. However remember the following:

1. Begin by clearing fractions, brackets and root
2. Rearrange the formula so that all the terms that contain the new subject are on one side of the equals sign and the rest on the other. Do not try to place the subject on the left-hand side if it comes more naturally on the right
3. If more than one term contains the subject, take it outside a bracket (i.e. factorise)
4. Divide both sides by the bracket, then simplify as far as possible.

EVALUATION

Make x the subject of the following equations.

1. $x(a - b) = b(c - x)$
2. $\sqrt{x^2 - a^2} = b$
3. $(ax - b)(bx + a) = (bx^2 + a)a$

8.3 Types of Variation such as: Direct, Inverse, Joint and Partial

Direct Variation

If a person buys some packets of sugar, the total cost is proportional to the number of packets bought

The cost of 2 packets at Nx per packet is $N2x$

The cost of 3 packets at Nx per packet is $N3x$.

The cost of n packets at Nx per packet is Nnx .

Thus, the ratio of total cost to number of packets is the same for any number of packets bought.

These are both examples of direct variation, or direct proportion. In the first example, the cost, C , varies directly with the number of packets, n .

Example 1

If 1 packet of sugar costs x naira what will be the cost of 20 packets?

Cost varies directly with the number of packets bought.

Cost of 1 packet = x naira

Cost of 20 packets = $20 \times n$ naira
= $20x$ naira

Example 2

If $C \propto n$ and $C = 5$ when $n = 20$, find the formula connecting C and n .

$C \propto n$ means $\frac{C}{n}$ is constant.

Let this constant be k .

Then, $\frac{C}{n} = k$

Or $C = kn$

$C = 5$ when $n = 20$

Hence $5 = k \times 20$

$$k = \frac{1}{4}$$

thus, $C = \frac{1}{4}n$ is the formula which connects C and n .

a formula such as $C = \frac{1}{4}n$ is often known as a relationship between the variables C and n .

Example 3

If $M \propto L$ and $M = 6$ when $L = 2$, find

- a. The relationship between M and L ,
- b. The value of L when $M = 15$.

- a. If $M \propto L$, then $M = kL$ when k is a constant.

$$M = kL$$

When $M = 6$, $L = 2$

Thus, $6 = k \times 2$

$$K = 3$$

Therefore $M = 3L$ is the relationship between M and L .

- b. $M = 3L$ and $M = 15$,

$$\text{Thus } 15 = 3L$$

$$L = 5$$

EVALUATION

1. If $P \propto Q$ and $P = 4.5$ when $Q = 12$, find
 - a. The relationship between P and Q
 - b. P when $Q = 16$
 - c. Q when $P = 2.4$

8.4 Inverse Variation

a. 5 equal sectors,

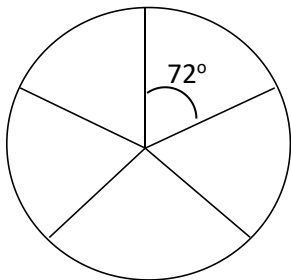


Fig 1

b. 12 equal sectors.

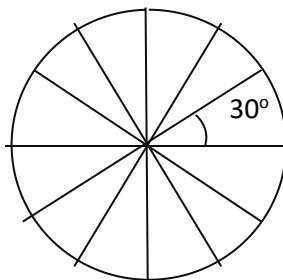


Fig 2

In fig 1, there are 5 equal sectors in the circle. The angle of each sector is 72° .

In fig 2, there are 12 equal sectors in the circle. The angle of each sector is 30° .

If there are 18 sectors, the angle of each sector would be 20° .

Therefore, the greater the number of sectors, the smaller the angle of each sector.

Similarly, if a car travels a certain distance, the greater its average speed, the less time it will take.

These are both examples of inverse variation, or inverse proportion. Sometimes known as indirect variation. In the first example, the size of the angle, θ , varies inversely with the number of sectors, n . In the second, the time taken, T , is inversely proportional to the average speed, S . These statements are written:

$$\theta \propto \frac{1}{n} \quad T \propto \frac{1}{S}$$

Example 1

If $\theta \propto \frac{1}{n}$ and $\theta = 72$ when $n = 5$, find

- a. θ when $n = 12$
- b. n when $\theta = 8$

First: find the relation between θ and n

$\theta \propto \frac{1}{n}$ means $\theta = \frac{k}{n}$ where k is a constant.

$$\theta = \frac{K}{n}$$

When $\theta = 72$, $n = 5$

Thus, $72 = \frac{K}{5}$

$$K = 5 \times 72 = 360$$

Thus, $\theta = \frac{360}{n}$

$$\text{a. } \theta = \frac{360}{n}$$

When $n = 12$,

$$\theta = \frac{360}{12} = 30$$

$$\text{b. If } \theta = \frac{360}{n} \text{ then } n = \frac{360}{\theta}.$$

When $\theta = 8$,

$$n = \frac{360}{8} = 45$$

8.5 Joint Variation

The mass of a sheet of metal is proportional to both the area and the thickness of the metal. Therefore $M \propto At$ (where M , A and t are the mass, area and thickness). This is an example of joint variation. The mass varies jointly with the area and thickness.

Example 1

Y varies inversely as X^2 and X varies directly as Z^2 . Find the relationship between Y and Z, given C is a constant.

From the first sentence:

$$Y \propto \frac{1}{X^2} \text{ and } X \propto Z^2$$

$$\text{Or } Y = \frac{A}{X^2} \text{ and } X = BZ^2$$

Where A and B are constants.

Substituting BZ^2 for X in $Y = \frac{A}{X^2}$

$$Y = \frac{A}{(BZ^2)^2} = \frac{A}{B^2Z^4}$$

$$\text{Or } Y = \frac{C}{Z^4} \text{ where C is a constant} = \left(\frac{A}{B^2}\right)$$

Thus Y varies inversely as Z^4 .

(Alternatively, $YZ^4 = C$)

EVALUATION

1. A rectangle has a constant area, A. its length is l and its breadth is b.
 - a. Write a formula for l in terms of A and b
 - b. Write a formula for b in terms of A and l.
 - c. Does l vary inversely or directly with b?

8.6 Partial Variation

When a tailor makes a dress, the total cost depends on two things: first the cost of the cloth; secondly the amount of the time it takes to make the dress. The cost of the cloth is constant, but the time taken to make the dress can vary. A simple dress will take a short time to make; a dress with a difficult pattern will take a long time. This is an example of partial variation. The cost is partly constant and partly varies with the amount of time taken. In algebraic form, $C = a + kt$, where C is the cost, t is the time taken and a and k are constants.

Example 1

R is partly constant and partly varies with E. when $R = 350$, $E = 1,600$ and when $R = 730$, $E = 3,600$.

- b. Find the formula which connects R and E.
- c. Find R when $E = 1300$

- a. From the first sentence,

$R = c + KE$ where c and k are constants. Substituting the given values gives two equations.

$$530 = c + 1600k \quad (1)$$

$$730 = c + 3600k \quad (2)$$

These are simultaneous equations.

Subtract (1) from (2)

$$200 = 2000k$$

$$K = \frac{200}{2000} = \frac{1}{10}$$

Substituting in (1),

$$530 = c + 1600 \times \frac{1}{10}$$

$$530 = c + 160$$

$$\text{Thus, } c = 370$$

Thus, $R = 370 + \frac{1}{10}E$ is the required formula.

$$\text{b. } R = 370 + \frac{E}{10}$$

when $E = 1300$,

$$R = 370 + \frac{1300}{10}$$

$$= 370 + 130$$

$$= 500$$

EVALUATION

- The cost of a car service is partly constant and partly varies with time it takes to do the work. It cost N3, 500 for a $5\frac{1}{2}$ hour service and N2, 900 for a 4-hour service.
 - Find the formula connecting cost, NC with time, T hours
 - Hence find the cost of a $7\frac{1}{2}$ hour service.
- X is partly constant and partly varies as y. when $y = 2$, $x = 30$, and when $y = 6$, $x = 50$.
 - Find the relationship between x and y.
 - Find x when $y = 3$

GENERAL EVALUATION

- If a man cycles 15km in 1 hour, how far will he cycle in two hours if he keeps up the same rate?
- A piece of string is cut into n pieces of equal length l.
 - Does n vary directly or inversely with l?
- The mass of rice that each woman gets when sharing a sack varies inversely with the number of women. When there are 20 women, each gets 6kg of rice. If there are nine woman, how much does each get?

READING ASSIGNMENT

New General Mathematics SSS 1 pages 220 Exercise 18a 11 – 15

WEEKEND ASSIGNMENT

- $x \propto \frac{y}{z}$ when $y = 7$ and $z = 3$, $x = 42$.
- Find the relation between x, y and z A. $y = xz$ B. $x = y/z$ C. $x = 18y/z$ D. $y = 18xz$
 - Find x when $y = 5$ and $z = 9$ A. 20 B. 5 C. 2 D. 10
 - Which of the following give the correct expressions for inverse variation A. $x = ky$ B. $x = k/y$ C. $x = kyz$ D. $x = 1/y$
 - If $r \propto 1/T$ and $T = 8$ when $R = 4$, find the relationship between R and T A. $R = 32/T$ B. $R = 32T$ C. $R = T/32$ D. $32 = R$
 - If D varies inversely as T use the symbol $\propto$ to show a connection between d and t. A. $d \propto t$ B. $d \propto 1/t$ C. $d = 1/2t$ D. $d \propto = 1/t$

THEORY

- The number of bricks, b, that a man can carry varies inversely with the mass of each brick, m kg, find the relationship between b and m. hence find the number of $3\frac{1}{2}$ kg bricks that he can carry.
- If $x - 3$ is directly proportional to the square of y and $x = 5$ when $y = 2$, find x when $y = 6$